

Note on: *Constant Maturity Treasury Convexity Correction*

Mario Pucci (Banca IMI), IJTAF 17(8), 2014

Reading notes

May 24, 2026

Metadata

Title	Constant Maturity Treasury Convexity Correction
Author	Mario Pucci (Banca IMI SpA)
Published	International Journal of Theoretical and Applied Finance 17(8), 2014
Dates	Received 13 Sep 2013; revised 31 Mar 2014, 4 Jul 2014; accepted 20 Aug 2014
Source	ssrn:3387961
Local file	<code>papers/pucci_2014_cmt/ssrn3387961.pdf</code>
Tags	<code>cmt, convexity, default-risk, partial-filtration, risky-annuity, lognormal</code>
Date read	May 24, 2026

One-paragraph summary

The paper develops a pricing framework for derivatives whose payoff depends on a Constant Maturity Treasury (CMT) yield, taking default risk of the bond issuer into account. The key conceptual move is to identify R_t^{cmt} with a *risky forward swap rate with recovery* on a credit-risk-adjusted curve, so that it is a ratio of tradable assets and therefore a martingale under the corresponding annuity (swaption) measure. The naive risky swap rate has a singularity at default; the partial filtration / credit-risk-adjusted framework (Bielecki–Jeanblanc–Rutkowski, Jamshidian) sidesteps this. Three versions of a CMT-let payoff are priced (A: survival to payment date; B: survival to fixing; C: B plus pre-default value if default precedes fixing), and closed-form convexity corrections are obtained under a one-factor linear swap-rate model with lognormal R^{cmt} dynamics. In the no-default limit, the results collapse to Pelsser’s classical convexity correction.

1 Setup and notation

The paper distinguishes carefully between three discount-bond objects: risk-free $D_{t,T}$, risky $\widetilde{D}_{t,T}$ (zero at default), and *credit-risk-adjusted / pre-default* $\widehat{D}_{t,T}$ (the analytically convenient version that remains continuous through default). The risky annuity $\widehat{A}_t$ is the natural numeraire for R^{cmt} .

Paper	House	Meaning
R_t^{ytm}	R_t^{ytm}	Yield-to-maturity of the reference bond at t
R_t^{cmt}	R_t^{cmt}	Constant Maturity Treasury rate; risky par-coupon rate
$\widehat{R}_t^{swp}$	$\widehat{R}_t^{swp}$	Risky forward swap rate with recovery ($= R_t^{cmt}$ at T_s)
B_t	$P_t(\cdot)$	Reference bond price
D_{tT}	$D_{t,T}$	Risk-free DF
$\widetilde{D}_{tT}$	$\widetilde{D}_{t,T}$	Risky DF ($= 0$ for $t \geq \tau$)
$\widehat{D}_{tT}$	$\widehat{D}_{t,T}$	Pre-default / credit-risk-adjusted DF (continuous in t)
$\widehat{D}_{tT}^{(R)}$	—	Risky DF with deterministic recovery Rec at default
$A_t, \widehat{A}_t$	$A_t, \widehat{A}_t$	Risky annuity, credit-risk-adjusted annuity
N_t	N_t	Numerator process
τ	τ	Default time
Γ_t	Γ_t	Hazard process: $e^{-\Gamma_t} = \mathbb{P}^N[\tau > t \mathcal{F}_t]$
γ_t	γ_t	Default intensity, $\Gamma_t = \int_0^t \gamma_u du$ under (AC)
χ_t	$\chi_t = e^{\Gamma_{T_s} - \Gamma_t}$	Forward survival ratio
Rec	Rec	Recovery rate (deterministic, level-independent)
T_s, T_p	T_s, T_p	Fixing date, payment date
y_i	α_i	Year fractions $yf(T_{i-1}, T_i)$
K	K	Bond coupon rate
α, β_U	α, β_U	Linear-model coefficients in $G_U(x) = \alpha + \beta_U x$
σ	σ	CMT lognormal vol
DI, AC	DI, AC	Default independence; absolute continuity of F_t

2 Key equations

Bond price as function of YTM (compounded convention):

$$P_{T_s}() = \prod_{i=1}^n \frac{1}{1 + \alpha_i R_{T_s}^{ytm}} + K \sum_{i=1}^n \alpha_i \prod_{j=1}^i \frac{1}{1 + \alpha_j R_{T_s}^{ytm}}. \quad (1)$$

Says: same compounded form as in Pucci's T-Lock paper; the YTM is a non-linear function of the bond price.

Definition of R^{cmt} (par-coupon on the risky curve).

$$1 = R_{T_s}^{cmt} \widehat{A}_{T_s} + \widehat{D}_{T_s T_n} + \text{Rec}[\text{default leg}]_{T_s}. \quad (2)$$

Says: $R_{T_s}^{cmt}$ is the fixed rate that prices a risky coupon bond at par, with the three summands being coupon PV, notional PV, and the recovery default leg. Equivalent definitions on the credit-risk-adjusted curve sidestep the default-time discontinuity.

Link between YTM and CMT (Proposition 2.1):

$$R^{ytm}(B, K) \cong R^{cmt} + (K - R^{cmt}) - \frac{R^{cmt}}{1 - (1 + R^{cmt})^{-n}} (B - 1) \cong R^{cmt} + (K - R^{cmt}) - \frac{B - 1}{n}. \quad (3)$$

Says: first-order Taylor expansion of R^{ytm} around the “at par” point $(B, K) = (1, R^{cmt})$. The expansion linearises YTM in B and K , so CMY pricing on the risky curve reduces (to first order) to a CMT-on-risky-swap-rate problem.

Three CMT-let payoffs (Defs. 3.1–3.3, payment at $T_p \geq T_s$):

$$(A) \quad \mathbf{1}_{[\tau > T_p]} R_{T_s}^{ytm}, \quad (4)$$

$$(B) \quad \mathbf{1}_{[\tau > T_s]} R_{T_s}^{ytm}, \quad (5)$$

$$(C) \quad \mathbf{1}_{[\tau > T_s]} R_{T_s}^{ytm} + \mathbf{1}_{[\tau \leq T_s]} R_{\tau-}^{ytm}. \quad (6)$$

Says: (A) is a survival-contingent payoff (no payment if default occurs in $[T_s, T_p]$); (B) is contingent on survival to the fixing date only; (C) is non-contingent: if default occurs before fixing, pays the last available fixing. NPV ordering: $\text{NPV}^{(C)} \geq \text{NPV}^{(B)} \geq \text{NPV}^{(A)}$.

Convexity correction (definition):

$$CC^{(i)} \equiv \frac{\text{NPV}^{(i)}}{D \cdot F} - 1, \quad i \in \{A, B, C\}, \quad (7)$$

where F is the forward of the rate and D the natural discount. *Says:* the convexity correction is what you add to the “forward rate times discount” deterministic value to obtain the true risk-neutral price.

One-factor linear swap-rate model (Sec. 5):

$$\frac{\widehat{D}_{T_s U}}{\widehat{A}_{T_s}} = G_U(R_{T_s}^{\text{cmt}}) = \alpha + \beta_U R_{T_s}^{\text{cmt}}, \quad (8)$$

for all $U \in \{T_p, T_1, \dots, T_n\}$, with

$$\alpha = \frac{1}{\sum_{i=1}^n \alpha_i}, \quad \beta_U = \frac{\chi_0 \widehat{D}_{0U} / \widehat{A}_0 - \alpha}{R_0^{\text{cmt}}}. \quad (9)$$

Says: the ratio of risky DF to risky annuity is modelled as a linear function of R^{cmt} . Calibration is forced by $R_0^{\text{cmt}} \rightarrow 0$ limit (for α) and martingale property under the annuity measure (for β_U).

Lognormal CMT dynamics:

$$\frac{dR_t^{\text{cmt}}}{R_t^{\text{cmt}}} = \sigma dW_t^{\widehat{A}}. \quad (10)$$

Says: R^{cmt} is lognormal under the annuity measure, with volatility σ . Combined with $W^{\widehat{A}}$ being a Brownian motion under the enlarged filtration $\mathcal{G}$, this gives closed-form expectations.

Closed-form convexity corrections (Proposition 6.1, under DI + lognormal; AC + constant intensity γ for (C)):

$$CC_0^{(A)} = CC_0^{(B)} = (e^{\sigma^2 T_s} - 1) \left(1 - \frac{\alpha \widehat{A}_0}{\chi_0 \widehat{D}_{0T_p}} \right), \quad (11)$$

$$CC_0^{(C)} = (1 + CC_0^{(B)}) \frac{\widehat{D}_{0T_s}}{D_{0T_s}} - 1 + \frac{\alpha \widehat{A}_0}{\chi_0 \widehat{D}_{0T_p}} (1 - e^{-\gamma T_s}) + \left(1 - \frac{\alpha \widehat{A}_0}{\chi_0 \widehat{D}_{0T_p}} \right) \frac{e^{(\sigma^2 - \gamma) T_s} - 1}{\sigma^2 - \gamma} \gamma. \quad (12)$$

Says: clean closed forms. $CC^{(A)} = CC^{(B)}$ holds because the survival difference between the two payoffs is exactly absorbed by the credit-risk-adjusted DF ratio in the discount. $CC^{(C)}$ has three distinct contributions: rate convexity (B), default-driven jump to pre-default value, and a γ -weighted integral. All vanish at $\sigma = 0$ (verified explicitly in the paper).

No-default limit (Sec. 6, $\gamma \rightarrow 0$):

$$CC_0^{(A)} = CC_0^{(B)} = CC_0^{(C)} \longrightarrow (e^{\sigma^2 T_s} - 1) \left(1 - \frac{\alpha A_0}{D_{0T_p}} \right), \quad (13)$$

Says: recovers the classical Pelsser (2001) convexity correction for a CMS-let under the linear swap-rate model.

3 Derivations & commentary

Commentary

The key structural move is identifying R^{cmt} with the risky forward swap rate with recovery: in the appendix Pucci shows $R_{T_s}^{\text{cmt}} = \hat{R}_{T_s}^{\text{sup}}$ at fixing. This buys two things: (i) R^{cmt} is a ratio of tradable assets, hence a martingale under the credit-risk-adjusted annuity measure $\hat{A}$; (ii) the machinery from CMS pricing (Hagan, Hunt–Kennedy linear swap-rate model) transfers almost verbatim, with $\hat{D}$ and $\hat{A}$ in place of D and A .

Commentary

The three payoff variants (A/B/C) matter operationally because they correspond to different contract designs. (A) and (B) are commonly embedded in survival-contingent structures (e.g. a swap that cancels on credit event). (C) is the building block for non-contingent CMT-linked notes. The ordering $\text{NPV}^{(C)} \geq \text{NPV}^{(B)} \geq \text{NPV}^{(A)}$ is structural, not assumption-dependent.

Commentary

The Taylor link (3) is a small but useful result on its own: it says that to first order in $(B - 1)$ and $(K - R^{\text{cmt}})$, the non-linear YTM equals an affine function of CMT. So once you can price CMT-contingent payoffs, you can approximate CMY-contingent payoffs by the same formula plus a known linear correction. The expansion deteriorates for bonds far from par; for ARM-mortgage indexing applications (the paper’s motivating use case) the bonds are near par and the approximation is tight.

Commentary

The $CC^{(C)}$ formula has a special structure worth noting: the $\gamma \rightarrow \sigma^2$ limit is regular (the $(\sigma^2 - \gamma)^{-1}$ factor cancels against the exponential), and the $\gamma \rightarrow 0$ limit collapses all three corrections to Pelsser. So the closed forms degrade smoothly as default risk disappears — a desirable property for a risky CMT framework.

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

Quantities the paper defines:

- Bond price as function of YTM (eq. 1), and its inverse $R^{\text{ytm}}(B, K)$ defined implicitly.
- $R_{T_s}^{\text{cmt}}$ as the risky par-coupon rate defined by (2).
- The YTM–CMT link (3) (first-order Taylor).

- Three CMT-let payoffs $\text{NPV}^{(A,B,C)}$ given by (4)–(6).
- Linear swap-rate model coefficients α, β_U from (9).
- Closed-form convexity corrections $CC^{(A)}, CC^{(B)}, CC^{(C)}$ in (11)–(12).

Canonical numerical case: the paper provides no worked numerical example in the body. Suggested test bench: a 10Y CMT-let on a par bond with $T_s = 5\text{Y}$, $T_p = T_s + 6\text{M}$, annual coupon (so $n = 10$, $\alpha_i = 1$), flat hazard γ across $\{0\%, 1\%, 3\%\}$, lognormal vol $\sigma \in \{10\%, 20\%\}$, flat risk-free curve at 2%.

Expected numerical values: no quantitative benchmarks printed in the paper. Validation is by closed-form vs. Monte Carlo simulation of the lognormal dynamics, plus limiting-case identities below.

Equation $\leftrightarrow$ quantity map:

Paper eq.	Symbol	Quantity
(1)	$P_{T_s}(R^{ytm})$	bond price as function of YTM
(2)	$R_{T_s}^{cmt}$	risky par-coupon rate
(3)	$R^{ytm}(B, K)$	first-order Taylor link to R^{cmt}
(4)–(6)	$\text{NPV}^{(A,B,C)}$	three CMT-let payoff variants
(8)–(9)	α, β_U	linear swap-rate model coefficients
(11)	$CC^{(A)} = CC^{(B)}$	closed-form CC for (A) and (B)
(12)	$CC^{(C)}$	closed-form CC for (C) under AC + const. γ

Invariants and identities to verify:

1. *YTM-CMT consistency:* at $B = 1$ and $K = R^{cmt}$, $R^{ytm}(B, K) = R^{cmt}$ exactly. (Pin point of the Taylor expansion.)
2. *$CC^{(A)} = CC^{(B)}$ identity:* the equality in (11) should hold for any valid set of inputs under DI, not just at the canonical case.
3. *Vanishing at zero vol:* $CC^{(A)} = CC^{(B)} = 0$ at $\sigma = 0$. $CC^{(C)}$ should also vanish identically at $\sigma = 0$ under DI + AC + constant γ ; this is verified algebraically in the paper.
4. *No-default limit:* as $\gamma \rightarrow 0$ (i.e. $\hat{A} \rightarrow A$, $\hat{D} \rightarrow D$, $\chi \rightarrow 1$), all three corrections collapse to the Pelsser single-curve form $(e^{\sigma^2 T_s} - 1)(1 - \alpha A_0 / D_{0T_p})$.
5. *NPV ordering:* $\text{NPV}^{(C)} \geq \text{NPV}^{(B)} \geq \text{NPV}^{(A)}$ for any feasible inputs.
6. *Regular limit at $\gamma = \sigma^2$:* $CC^{(C)}$ does not blow up at $\gamma = \sigma^2$; the integral in (12) has a removable singularity and the limiting value can be evaluated via L'Hôpital.
7. *Closed form vs. Monte Carlo:* simulate R^{cmt} under (10) together with a doubly-stochastic default time of intensity γ ; the MC estimator of $CC^{(C)}$ should match (12) within statistical error.

5 Glossary of symbols (paper's notation)

$T_s, T_1, \dots, T_n$	Payment schedule of the reference bond
T_p	Payment date of the CMT-let, $T_p \geq T_s$
y_i	Year fractions $yf(T_{i-1}, T_i)$
K	Coupon rate of the reference bond
B_t	Reference bond price at t
R_t^{ytm}	Yield-to-maturity at t
$R_{T_s}^{cmt}$	Constant Maturity Treasury rate at T_s
$\hat{R}_t^{swp}$	Risky forward swap rate with recovery (equals R^{cmt} at T_s)
D_{tT}	Risk-free discount bond price
$\tilde{D}_{tT}$	Risky discount bond ($= 0$ for $t \geq \tau$)
$\hat{D}_{tT}$	Pre-default (credit-risk-adjusted) risky DF, continuous in t
$\hat{D}_{tT}^{(R)}$	Risky DF with deterministic recovery Rec on default
$A_t, \hat{A}_t$	Risky annuity, credit-risk-adjusted annuity
N_t	Numeraire process (paper picks $N = \hat{A}$)
τ	Default time
Γ_t	Hazard process; $e^{-\Gamma_t} = \mathbb{P}^N[\tau > t \mathcal{F}_t]$
γ_t	Default intensity (under AC, $\Gamma_t = \int_0^t \gamma_u du$)
χ_t	$e^{\Gamma_{T_s} - \Gamma_t}$, forward survival ratio
Rec	Deterministic recovery rate
$\mathcal{F}_t, \mathcal{G}_t$	Default-free / market filtration
σ	Volatility of R^{cmt} under lognormal dynamics
α, β_U	Coefficients of the linear swap-rate model
$CC^{(i)}$	Convexity correction for payoff $i \in \{A, B, C\}$
DI	“Default Independence” axiom: $\tau \perp \mathcal{F}$
AC	“Absolute Continuity” axiom: $F_t = 1 - e^{-\Gamma_t}$ is AC

TODO / Open questions

- Build a canonical numerical test (10Y CMT-let, 5Y fixing, flat hazard $\gamma \in \{0\%, 1\%, 3\%\}$, $\sigma \in \{10\%, 20\%\}$) and produce a small table of $CC^{(A,B,C)}$ to anchor the closed forms.
- Monte Carlo cross-check: simulate R^{cmt} under (10) jointly with a Cox-process default time, estimate $NPV^{(C)}$, compare to (12).
- Verify the $\gamma \rightarrow \sigma^2$ regularity of $CC^{(C)}$ numerically; evaluate by L'Hôpital and compare to a small- ϵ direct evaluation.
- Code the YTM–CMT Taylor link (3) and quantify the second-order error for bonds at $\pm 5\%$ off par.
- Smile extension: the paper sketches a static-replication approach for incorporating R^{cmt} smile via the $\mathbb{E}[(R_{T_s}^{cmt})^2]$ term; not in scope for first-pass validation.